MATH 152
Mrs. Bonny Tighe

QUIZ 6
25 points
7.8,8.1,8.2

NAME Solutions

TA_____ Wed. 10/16/2019

1pt 1. The area under a curve can be found if the improper definite integral is Convergent

2. Describe the two types of improper integrals.

1pt 1- a limit of integration is an infinity

2- a limit of integration is undefined

3. State the Comparison Test for Improper Integrals

2pts Suppose f and g are continuous functions with $f(x) \geq g(x) \geq 0$ for $x \geq a$. a) If $\int_a^\infty f(x)\,dx$ is convergent, then $\int_a^\infty g(x)\,dx$ is convergent b) If $\int_a^\infty g(x)\,dx$ is divergent, then $\int_a^\infty f(x)\,dx$ is also divergent

4. Determine whether each integral is convergent or divergent. Evaluate those that are convergent.

4pts a) $\int_1^9 \frac{1}{\sqrt[3]{x-1}}\,dx$

$$\lim_{t\to 1^+} \int_t^9 (x-1)^{-1/3}\,dx$$

$$\lim_{t\to 1^+} \frac{3}{2}(x-1)^{2/3}\Big|_t^9$$

$$\lim_{t\to 1^+} \left[\frac{3}{2}(8)^{2/3} - \frac{3}{2}(t-1)\right]$$

$$\frac{3}{2}(4) - \frac{3}{2}(0)$$

$$6$$

so Convergent

4pts b) $\int_1^\infty \frac{e^{-\sqrt{x}}}{\sqrt{x}}\,dx$

$u = -\sqrt{x}$

$du = -\frac{1}{2\sqrt{x}}\,dx$

$-2\sqrt{x}\,dx = dx$

$$\lim_{t\to\infty} \int_1^t -2e^u\,du =$$

$$\lim_{t\to\infty} -2e^u\Big| = \lim_{t\to\infty} -2e^{-\sqrt{x}}\Big|_1^t$$

$$\lim_{t\to\infty} \left[-2e^{-\sqrt{t}} - (-2e^{\sqrt{1}})\right]$$

$$-2e^{-\infty} + 2e^{-1} = \frac{2}{e}$$

so Convergent

5. Use the comparison test to determine whether the integrals diverge or converge.

3pts $\int_3^\infty \frac{x}{\sqrt{x^3-5x}}dx$ Compare to $\int_3^\infty \frac{1}{\sqrt{x}}dx$ which is divergent

$p = 1/2 \leq 1$

$\frac{1}{\sqrt{x}} \leq \frac{x}{\sqrt{x^3-5x}}$

$\sqrt{x^3-5x} \leq x\sqrt{x}$

$x^3-5x \leq x^3$

$0 \leq 5x$

which is true for $x \geq 3$

So $\int_3^\infty \frac{x}{\sqrt{x^3-5x}}dx$ is also divergent

6. Find the length of the arc of the curve $y = \frac{x^3}{3} + \frac{1}{4x}$, $1 \leq x \leq 2$

5pts

$y = \frac{1}{3}x^3 + \frac{1}{4}x^{-1}$

$L = \int_1^2 \sqrt{1+(dy/dx)^2}\,dx$

$dy/dx = x^2 - \frac{1}{4}x^{-2}$

$x^2 - \frac{1}{4x^2}$

$(dy/dx)^2 = x^4 - 2(x^2)(\frac{1}{4x^2}) + \frac{1}{16x^4}$

$= x^4 - \frac{1}{2} + \frac{1}{16x^4}$

$L = \int_1^2 \sqrt{1+x^4-\frac{1}{2}+\frac{1}{16x^4}}\,dx = \int_1^2 \sqrt{x^4+\frac{1}{2}+\frac{1}{16x^4}}\,dx$

one of them

$L = \int_1^2 \sqrt{(x^2+\frac{1}{4x^2})^2}\,dx = \int_1^2 (x^2+\frac{1}{4}x^{-2})\,dx$

$\frac{1}{3}x^3 + \frac{1}{4}\cdot\frac{1}{-1}x^{-1}\Big|_1^2 = \frac{1}{3}x^3 - \frac{1}{4x}\Big|_1^2$

$(\frac{1}{3}(8) - \frac{1}{8}) - (\frac{1}{3} - \frac{1}{4}) = \frac{8}{3} - \frac{1}{8} - \frac{1}{3} + \frac{1}{4} = \frac{7}{3} + \frac{1}{8}$

$x = r$

5pts 7. The given curve is rotated about the y-axis. Find the area of the resulting surface.

$x = \sqrt{4-y} \quad 1 \le y \le 3 \qquad S = 2\pi \int_a^b radius\ arclength$

$x = (4-y)^{1/2}$

$dx/dy = \frac{1}{2}(4-y)^{-1/2}(-1)$

$= \frac{-1}{2\sqrt{4-y}}$

$(dx/dy)^2 = \frac{1}{4(4-y)}$

$S = \int_1^3 \sqrt{4-y}\sqrt{1+(dx/dy)^2}\,dy$

$S = \int_1^3 \sqrt{4-y}\sqrt{1+\frac{1}{4(4-y)}}\,dy$

$S = \int_1^3 \sqrt{4-y}\sqrt{\frac{16-4y+1}{4(4-y)}}\,dy$

$S = \int_1^3 \cancel{\sqrt{4-y}}\,\frac{\sqrt{17-4y}}{2\cancel{\sqrt{4-y}}}\,dy$

$S = \int_1^3 \frac{1}{2}(17-4y)^{1/2}\,dy = \frac{1}{2}\cdot\frac{-1}{3/2}(17-4y)^{3/2}\Big|_1^3$

$-\frac{1}{3}(17-4y)^{3/2}\Big|_1^3 = -\frac{1}{3}\left[(17-12)^{3/2} - (17-4)^{3/2}\right]$

$-\frac{1}{3}\left[5^{3/2} - 13^{3/2}\right]$ (positive)

MATH 152 **QUIZ 6A** NAME Solutions

Mrs. Bonny Tighe 25 points

7.8,8.1,8.2 TA_____ Wed. 10/16/2019

2pts 1. State the Comparison Test for Improper Integrals

Suppose f and g are continuous functions with $f(x) \geq g(x) \geq 0$ for $x \geq a$. a) if $\int_a^\infty f(x)\,dx$ is convergent, then $\int_a^\infty g(x)\,dx$ is also convergent b) if $\int_a^\infty g(x)\,dx$ is divergent, then $\int_a^\infty f(x)\,dx$ is also divergent

1pt 2. The area under a curve can be found if the improper definite integral is Convergent

1pt 3. Describe the two types of improper integrals.

1- when a limit of integration is ∞

2- when a limit of integration is undefined

4pts 4. Determine whether each integral is convergent or divergent. Evaluate those that are convergent.

a) $\int_1^\infty \frac{e^{1/x}}{x^2}\,dx$

$u = \frac{1}{x}$

$du = -\frac{1}{x^2}\,dx$

$\lim_{t\to\infty} \int_1^t \frac{e^{1/x}}{x^2}\,dx$

$-e^u\,du = -e^u$

$\lim_{t\to\infty} -e^{1/x}\Big|_1^t =$

$\lim_{t\to\infty} \left(-e^{1/t} - (-e^{1/1})\right)$

$-e^0 + e^1 = -1 + e$

So Convergent

4pts b) $\cancel{\int_{-1}^1 \frac{1}{\sqrt{x+1}}\,dx}$ $\int_0^1 \frac{1}{\sqrt{1-x^2}}\,dx$

$\lim_{t\to 1^-} \int_0^t \frac{1}{\sqrt{1-x^2}}\,dx$

$\lim_{t\to 1^-} \left[\sin^{-1} x\right]_0^t$

$\lim_{t\to 1^-} \left[\sin^{-1} t - \sin^{-1} 0\right]$

$\sin^{-1}(1) - 0$

$\pi/2$

So convergent

3pts 5. Use the comparison test to determine whether the integrals diverge or converge.

$$\int_1^\infty \frac{x}{\sqrt{x^5+5x}}dx$$

Compare to $\int_1^\infty \frac{1}{x^{3/2}}dx$ which is convergent $p = 3/2 > 1$

$$\frac{1}{x^{3/2}} \geq \frac{x}{\sqrt{x^5+5x}}$$

$$\sqrt{x^5+5x} \geq x^{5/2}$$

$$x^5+5x \geq x^5$$

$5x \geq 0$ true for $x \geq 1$

So $\int_1^\infty \frac{x}{\sqrt{x^5+5x}}dx$ is also convergent

5pts 6. Find the length of the arc of the curve $x = \ln(y^2-1)$, $2 \leq y \leq 4$

$$L = \int_2^4 \sqrt{1+(dx/dy)^2}\,dy$$

$$dx/dy = \frac{1}{y^2-1}(2y)$$

$$(dx/dy)^2 = \frac{4y^2}{(y^2-1)^2}$$

$$L = \int_2^4 \sqrt{1+\frac{4y^2}{(y^2-1)^2}}\,dy$$

$$L = \int_2^4 \sqrt{\frac{(y^2-1)^2+4y^2}{(y^2-1)^2}}\,dy = \int_2^4 \frac{\sqrt{y^4-2y^2+1+4y^2}}{y^2-1}\,dy$$

$$\int_2^4 \frac{\sqrt{y^4+2y^2+1}}{y^2-1}\,dy = \int_2^4 \frac{\sqrt{(y^2+1)^2}}{y^2-1}\,dy = \int_2^4 \frac{y^2+1}{y^2-1}\,dy$$

$$\begin{array}{r} 1 \\ y^2-1 \overline{\big) y^2+1} \\ -(y^2-1) \\ \hline 2 \end{array}$$

$$\int_2^4 \left(1+\frac{2}{y^2-1}\right)dy$$

$$\frac{A}{y+1}+\frac{B}{y-1} = \frac{2}{y^2-1}$$

$$A(y-1)+B(y+1) = 2$$

$$B = 1$$

$$\int_2^4 1+\frac{1}{y-1}-\frac{1}{y+1}\,dy$$

$$y+\ln\left|\frac{y-1}{y+1}\right|\Big]_2^4 = \left(4+\ln\left|\frac{3}{5}\right|\right)-\left(2+\ln\left|\frac{1}{3}\right|\right)$$

$$2+\ln|3/5|-\ln|1/3| = 2+\ln\left|\frac{9}{5}\right|$$

$r = y$

5pts 7. The given curve is rotated about the x-axis. Find the area of the resulting surface.

$x = \cancel{\frac{y^3}{3} + \frac{1}{4y}}, \quad \cancel{1 \le y \le 2}$ $\qquad S = 2\pi \int_a^b radius\ arclength$

$y = \sqrt{1+e^x} \qquad 0 \le x \le 1$

$dy/dx = \frac{1}{2}(1+e^x)^{-1/2} e^x$

$\frac{e^x}{2\sqrt{1+e^x}}$

$(dy/dx)^2 = \frac{e^{2y}}{4(1+e^x)}$

$$S = 2\pi \int_0^1 \sqrt{1+e^x}\sqrt{1+(dy/dx)^2}\,dx$$

$$2\pi \int_0^1 \sqrt{1+e^x}\sqrt{1+\frac{e^{2x}}{4(1+e^x)}}\,dx$$

$$2\pi \int_0^1 \sqrt{1+e^x}\sqrt{\frac{4+4e^x+e^{2x}}{4(1+e^x)}}\,dx$$

$$2\pi \int_0^1 \cancel{\sqrt{1+e^x}}\,\frac{\sqrt{(e^x+2)^2}}{2\cancel{\sqrt{1+e^x}}}\,dx = \pi \int_0^1 (e^x+2)\,dx$$

$$\pi\left[e^x+2x\right]_0^1 = \pi\left[(e^1+2)-(e^0+0)\right] =$$

$$\pi\left[e+1\right]$$